

Week 3

Category / Ratio	2023	2022	2021
PROFITABILITY			
Vertical Analysis (% of Net Sales)			
Sales	100.0%	100.0%	100.0%
Cost of Goods Sold	69.0%	70.0%	73.5%
Gross Income (Profit)	31.0%	30.0%	26.5%
Bad Debt Expense	0.3%	0.1%	1.4%
General & Admin Expense	3.1%	3.5%	3.8%
Salaries	13.5%	13.9%	14.3%
Selling Expense	4.0%	3.8%	3.9%
Depreciation	1.3%	1.1%	1.4%
Income from Operations	↑ 8.7%	7.5%	1.9%
Other Income	1.4%	2.3%	2.0%
Interest Expense	1.3%	0.9%	1.3%
Income Before Tax	8.8%	8.9%	2.6%
Income After Tax (Net Profit Margin)	↑ 4.7%	4.8%	1.3%
Return on Average Equity (ROE)	18.9%	16.5%	N/A 4.2%
EFFICIENCY			
Age of Receivables	72.1 days	70.7 days	71.7 days
Age of Inventory (using COGS)	↑ 117.6 days	111.7 days	120.6 days
Age of Payables (using Purchases)*	↑ 52.3 days	↑ 30.8 days	23.3 days
LIQUIDITY			
Current Ratio	1.75 : 1	2.06 : 1	2.02 : 1

Not A Big focus

most recent

Now we need more loans

1. Profitability Ratios

These measure a company's ability to generate income relative to revenue, assets, or equity.

- o **Gross Profit Margin:** Measures percentage of revenue remaining after deducting direct production costs.

$$\text{Gross Profit Margin} = \frac{\text{Gross Profit}}{\text{Net Sales}}$$

- o **Net Profit Margin:** Percentage of revenue left after paying all operating expenses, interest, and taxes.

$$\text{Net Profit Margin} = \frac{\text{Net Income}}{\text{Net Sales}}$$

- o **Return on Equity (ROE):** Indicates how effectively the company uses shareholders' capital to generate profit.

$$\text{ROE} = \frac{\text{Net Income}}{\text{Shareholders' Equity}}$$

2. Liquidity Ratios

These assess a firm's ability to cover its short-term debt obligations without raising outside capital.

- o **Current Ratio:** Measures whether short-term assets can cover short-term liabilities.

$$\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}}$$

- o **Quick (Acid-Test) Ratio:** A stricter liquidity test that excludes inventory (less liquid asset).

$$\text{Quick Ratio} = \frac{\text{Cash} + \text{Accounts Receivable} + \text{Short-Term Investments}}{\text{Current Liabilities}}$$

3. Efficiency & Activity Ratios

These evaluate how productively a company manages its resources and working capital.

- o **Age of Inventory / Inventory Turnover:** Measures how quickly inventory is converted into sales.

$$\text{Age of Inventory (Days)} = \left(\frac{\text{Inventory}}{\text{COGS}} \right) \times 365$$

Age of Receivables (Collection Period): Measures average number of days taken to collect customer receivables.

$$\text{Age of Receivables (Days)} = \left(\frac{\text{Accounts Receivable}}{\text{Net Sales}} \right) \times 365$$

- o **Age of Payables:** Measures average number of days taken to pay trade suppliers.

$$\text{Age of Payables (Days)} = \left(\frac{\text{Accounts Payable}}{\text{Purchases}} \right) \times 365$$

4. Solvency & Leverage Ratios

These measure long-term financial stability and a firm's reliance on debt to finance operations.

- o **Debt-to-Equity (D/E):** Compares total debt liabilities against total owner equity.

$$\text{Debt-to-Equity} = \frac{\text{Total Liabilities}}{\text{Total Shareholders' Equity}}$$

- o **Interest Coverage Ratio:** Evaluates how easily earnings cover annual interest payments.

$$\text{Interest Coverage} = \frac{\text{EBIT}}{\text{Interest Expense}}$$

5. Growth Ratios

These track period-over-period percentage changes to evaluate company expansion rates.

- o **Sales Growth:** Percentage change in top-line revenue over a specified period.

$$\text{Sales Growth} = \frac{\text{Sales}_{\text{Current}} - \text{Sales}_{\text{Prior}}}{\text{Sales}_{\text{Prior}}} \times 100$$

- o **Net Income Growth:** Percentage change in net bottom-line earnings over a specified period.

38.6%

Current ratio

= current Assets $\div$ current liabilities

eg. 2:1

Acid test

= Cash + M/S + A/R $\div$ current liability

eg. 1:1

debt to equity

eg. 3:1

Net worth to total Assets

Eg. 25%

interest coverage

Eg. high as possible

Growth Cycle

38.6%
Sales growth

